Problema 1

$$V = \begin{pmatrix} U(1 - R^2/r^2)\cos\theta \\ -U(1 + R^2/r^2)\,\text{sen}\theta + \Gamma/2\pi r \\ 0 \end{pmatrix}$$

Para luego utilizar Bernoulli, hay que verificar que sea irrotacional:

$$\nabla \times \vec{V} = \begin{pmatrix} \frac{1}{r}\frac{\partial V_z}{\partial \theta}^{0} - \frac{\partial V_\theta}{\partial z}^{0} \\ \frac{\partial V_r}{\partial z}^{0} - \frac{\partial V_z}{\partial r}^{0} \\ \frac{1}{r}\frac{\partial (r V_\theta)}{\partial r} - \frac{1}{r}\frac{\partial V_r}{\partial \theta} \end{pmatrix}$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(-U(r + R^2/r)\,\text{sen}\theta + \Gamma/2\pi\right) - \frac{1}{r}\frac{\partial}{\partial \theta}\left(U(1 - R^2/r^2)\cos\theta\right)$$

$$-\frac{U}{r}(1 - R^2/r^2)\,\text{sen}\theta + \frac{U}{r}(1 - R^2/r^2)\,\text{sen}\theta = 0$$ ✓ Podemos usar Bernoulli

> Bernoulli entre un par de puntos a la misma altura:

$$(1)\quad \frac{P_\infty}{\gamma} + \frac{U_\infty^2}{2g} = \frac{P_{ro}}{\gamma} + \frac{V_{ro}^2}{2g}$$

En donde la velocidad del rotor es:

$$V_{ro}^2(r = R) = \left(U(1 - R^2/R^2)\cos\theta\right)^2 + \left(-U(1 + R^2/R^2)\,\text{sen}\theta + \Gamma/2\pi R\right)^2$$

$$V_{ro}^2(r = R) = \left(2U\,\text{sen}\theta + \Gamma/2\pi R\right)^2$$

$$(1)\quad P_{rotor} = \frac{\rho}{2g}\left(V_{ro}^2 - U_\infty^2\right) - P_\infty$$

$$P_{rotor} = \frac{\rho}{2g}\left[\left(2U\,\text{sen}\theta + \Gamma/2\pi R\right)^2 - U_\infty^2\right] - P_\infty$$

Fuerza: $d\vec{F} = -P\hat{n}dA = -P(Rd\theta dz)\cdot \hat{e}r$ → Habrá una fuerza en x e y.

$\hat{e}r = \cos\theta\,\hat{i} + \text{sen}\theta\,\hat{j}$

$$Fx = \int_0^L \int_0^{2\pi} \left(\frac{\rho}{2g}\left[\left(2U\,\text{sen}\theta + \Gamma/2\pi R\right)^2 - U_\infty^2 \right] - P_\infty \right) \cos\theta\, R\, d\theta\, dz$$

$$Fy = \int_0^L \int_0^{2\pi} \left(\frac{\rho}{2g}\left[\left(2U\,\text{sen}\theta + \Gamma/2\pi R\right)^2 - U_\infty^2 \right] - P_\infty \right) \text{sen}\theta\, R\, d\theta\, dz$$

Problema 2

Ecuación de continuidad para un flujo incompresible:

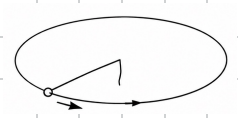
$$\frac{1}{r}\frac{\partial(r\,u_r)}{\partial r} + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_z}{\partial z} = 0$$

a) $u_z = 0$, $u_r = 0$

$$\frac{1}{r}\frac{\partial(r\,u_r)}{\partial r}^{0} + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta} + \frac{\partial u_z}{\partial z}^{0} = 0$$

$$\frac{\partial u_\theta}{\partial \theta} = 0 \quad , \quad u_\theta = C_1$$

$$V = \begin{pmatrix} u_r \\ u_\theta \\ u_z \end{pmatrix} = \begin{pmatrix} 0 \\ C_1 \\ 0 \end{pmatrix}$$



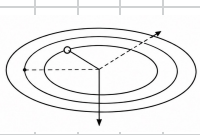
b) $u_z = 0$, $u_\theta = \frac{A}{r}(1 - e^{(-ar)^2})$

$$\frac{1}{r}\frac{\partial(r\,u_r)}{\partial r} + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta}^{0} + \frac{\partial u_z}{\partial z}^{0} = 0$$

$$\frac{\partial(r\,u_r)}{\partial r} = 0 \quad , \quad r\,u_r = C_2$$

$$u_r = C_2 / r$$

$$V = \begin{pmatrix} C_2/r \\ A(1 - e^{-(ar)^2})/r \\ 0 \end{pmatrix}$$



c) $u_z = 2az$, $u_\theta = \frac{A}{r}(1 - e^{(-ar)^2})$

$$\frac{1}{r}\frac{\partial(r\,u_r)}{\partial r} + \frac{1}{r}\frac{\partial u_\theta}{\partial \theta}^{0} + \frac{\partial u_z}{\partial z} = 0$$

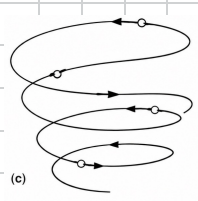
$$\frac{1}{r}\frac{\partial(r\,u_r)}{\partial r} + 2a = 0$$

$$\int d(r\,u_r) = -\int 2ar\,dr$$

$$r\,u_r = -ar^2 + C_3$$

$$u_r = -ar + \frac{C_3}{r}$$

$$V = \begin{pmatrix} -ar + C_3/r \\ A(1 - e^{-(ar)^2})/r \\ 0 \end{pmatrix}$$



Problema 3

N-S en z:

$$\rho\left(\frac{\partial v_z}{\partial t}+v_r\frac{\partial v_z}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta}+v_z\frac{\partial v_z}{\partial z}\right)=-\frac{\partial p}{\partial z}+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 v_z}{\partial \theta^2}+\frac{\partial^2 v_z}{\partial z^2}\right]+\rho g,$$

$$\rho\left(\frac{\partial v_z}{\partial t}+v_r\frac{\partial v_z}{\partial r}+\frac{v_\theta}{r}\frac{\partial v_z}{\partial \theta}+v_z\frac{\partial v_z}{\partial z}\right)=-\frac{\partial p}{\partial z}+\mu\left[\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)+\frac{1}{r^2}\frac{\partial^2 v_z}{\partial \theta^2}+\frac{\partial^2 v_z}{\partial z^2}\right]+\rho g,$$

régimen permanente

$v_r = v_\theta = 0$

$v_z(r)$

$$0=-\frac{\partial P}{\partial z}+\mu\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)+\rho g$$

Como está expuesto a la atmósfera, asumiremos que en todo el fluido no hay variación a lo largo de z $\quad \frac{\partial P}{\partial z}\approx 0$

$$\mu\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)+\rho g=0$$

$$\boxed{\frac{\partial}{\partial r}\left(r\frac{\partial v_z}{\partial r}\right)=-\frac{r\rho g}{\mu}}$$

Condiciones de borde:

- Como la barra está fija, en el borde se cumple la condición de no deslizamiento.

$$v_z(r=a)=0 \quad (1)$$

- La superficie libre expuesta a la atmósfera no opone resistencia de cortadura (enunciado).

$$\tau_{rz}(r=b)=0 \rightarrow \mu\left.\frac{dv_z}{dz}\right|_{r=b}=0 \rightarrow \left.\frac{dv_z}{dz}\right|_{r=b}=0 \quad (2)$$

Integramos:

$$\frac{\partial}{\partial r}\left(r\frac{\partial V_z}{\partial r}\right) = -\frac{r\rho g}{\mu}$$

$$\int d\left(r\frac{dV_z}{dr}\right) = -\int \frac{r\rho g}{\mu}\, dr$$

$$r\frac{dV_z}{dr} = -\frac{\rho g}{2\mu}r^2 + C_1$$

$$\int dV_z = \int\left(-\frac{\rho g}{2\mu}r + \frac{C_1}{r}\right)dr$$

$$V_z(r) = -\frac{\rho g}{4\mu}r^2 + C_1\ln(r) + C_2$$

Usamos las condiciones:

(2) $$\left.\frac{dV_z}{dr}\right|_{r=b} = -\frac{\rho g}{2\mu}b + \frac{C_1}{b} = 0 \quad , \quad C_1 = \frac{\rho g}{2\mu}b^2$$

(1) $$V_z(r=a) = -\frac{\rho g}{4\mu}a^2 + \frac{\rho g}{2\mu}b^2\cdot\ln a + C_2 = 0$$

$$C_2 = \frac{\rho g}{2\mu}\left(\frac{a^2}{2} - b^2\cdot\ln(a)\right)$$

↳ $$V_z(r) = \frac{\rho g}{2\mu}\left(\frac{a^2-r^2}{2} + b^2\cdot\ln\left(\frac{r}{a}\right)\right)$$ //